

Assessment Problem 3: Systems of Particles

Problem Statement

Consider a system of **5 particles** distributed in a Cartesian coordinate system (xOy).

- The coordinates are given in **centimeters (cm)**.
- The masses are given in **kilograms (kg)**.

Particle Data:

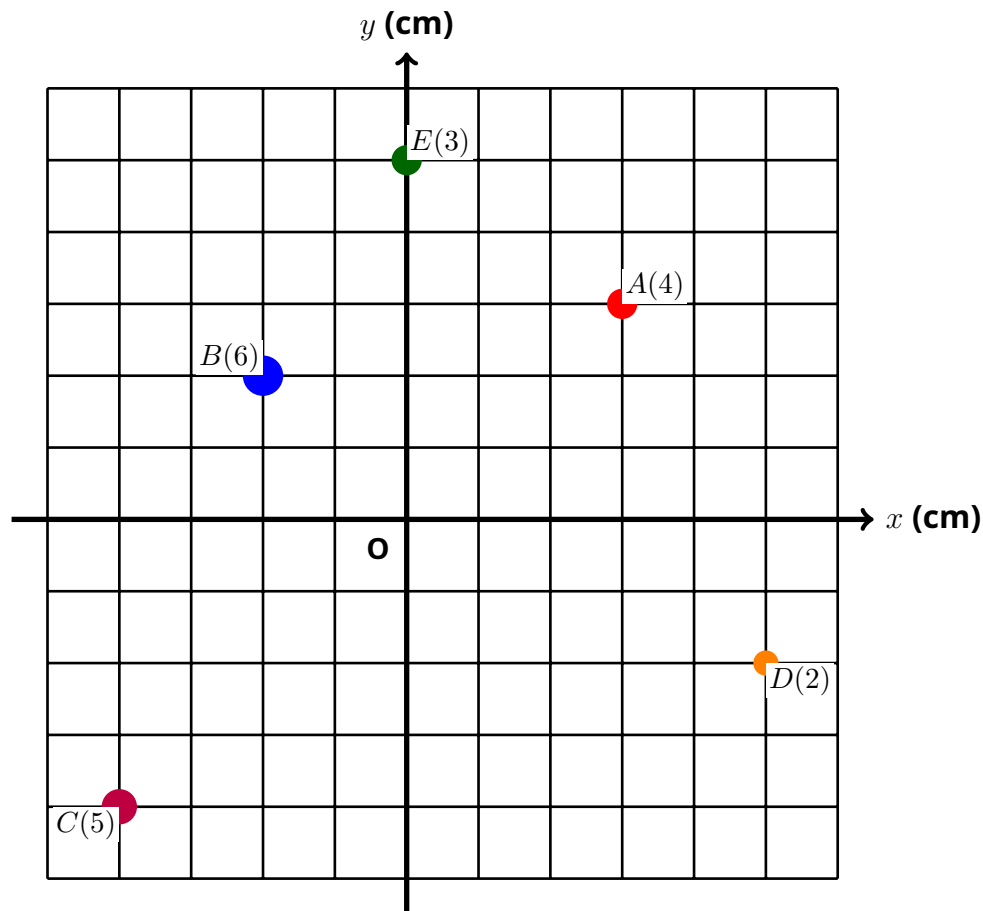
P1: 4 kg at $A(3, 3)$

P2: 6 kg at $B(-2, 2)$

P3: 5 kg at $C(-4, -4)$

P4: 2 kg at $D(5, -2)$

P5: 3 kg at $E(0, 5)$



Questions

Please solve on a separate sheet of paper.

1. **(1 point)** Calculate the **Total Mass** (M_{sys}) of the system.
2. **(3 points)** Determine the **coordinates** (x_G, y_G) of the Center of Mass. Show your work for both x and y calculations.
3. **(1 point) Analysis:** Based on your calculated coordinates, in which **Quadrant** is the Center of Mass located?